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Verifying Nondeterministic Sampling Algorithms via Distributional Invariants

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We present a Hoare-like verification framework for discrete probabilistic programs with *nondeterministic* choices, applied to two non-trivial case studies: the Nim game and an error-tolerant variant of Lumbroso’s Fast Dice Roller in which fair coin flips are replaced by adversarial choices biased by up to ϵ . Our framework extends inductive *distributional loop invariants* from the purely probabilistic setting to programs that combine randomness with nondeterminism, supporting both *existential* verification (some strategy achieves the goal) and *universal* verification (a property holds under all strategies of a chosen class). At the core of our development is a fine-grained landscape of program-level strategies—ranging from memoryless deterministic to history-dependent randomized—and an explicit translation theorem showing that program-level strategies precisely correspond to strategies on the program’s underlying Markov decision process. As a key intermediate result, we prove that *loop-counter* strategies are as expressive as full-memory strategies, yielding a more compact and intuitive program-level strategy that retains the full expressive power of full-memory strategies. Using this framework, we verify a winning strategy and a strategy-universal zero-sum property for the Nim game, and prove that the nondeterministic Fast Dice Roller produces a ϵ -uniform output distribution under *all* adversarial coin strategies.

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